PRESENTS

AURELIUS

CONSULTING & ANALYTICS CLUB

RECRUITMENT PROBLEM STATEMENT 2026

Consulting & Strategy Track

SOLARIS

CHALLENGE TITLE:

Case Study: Solaris Dynamics

1. Company Introduction

Solaris Dynamics Inc. is a Bengaluru-headquartered green-tech manufacturer and one of India's fastest-growing players in the sustainable energy storage sector. The company produces two flagship product lines: Titan Inverters, high-capacity industrial inverters targeted at infrastructure and power projects, and Eco-Battery Packs, modular energy storage units sold across commercial and retail segments.

Solaris operates manufacturing hubs across Bengaluru, Pune, and Hyderabad in India, with additional international facilities in Bonn, Duisburg, and Wuppertal, acquired as part of Solaris's 2017 European expansion into the renewable energy supply chain. Orders are fulfilled through a network of Standard and Smart (automated) warehouses and dispatched via third-party logistics partners including BlueDart, Delhivery, FedEx, and DB Schenker.

Solaris serves four customer markets — Retail, Transportation, Construction, and Other — with its Transportation segment anchored by marquee institutional clients such as NTPC Limited, Tata Power Renewables, Adani Green Energy, and Shapoorji Pallonji. These top five clients alone account for over 55% of Solaris's total revenue.

2. Business Case & The Real Problem

The Surface Story

Solaris's most recent internal review reports an overall On-Time Delivery (OTD) rate of 87.3% — comfortably above the company's stated 80% target. The operations team considers this a success. The board is satisfied.

The Problem Underneath

A closer look reveals that this headline number is masking a serious structural vulnerability.

The Transportation segment — Solaris's most valuable by far, generating ₹667 billion in order value — has an OTD rate of only 77.7%, below the 80% target. This is the segment that houses Solaris's largest institutional accounts, for whom a pattern of late deliveries is not an inconvenience — it is grounds for contract termination.

At the same time, the two product lines are telling very different stories. Eco-Battery Packs deliver on time 94.5% of the time. Titan Inverters deliver on time just 80.8% of the time — barely at target — and are responsible for 2,574 fulfilment shortfalls where fewer units were delivered than ordered. Given that Titan Inverters are priced significantly lower (avg ₹488/unit) than Eco-Batteries (avg ₹733/unit), the question of whether Titan Inverter operations are worth their cost to maintain is becoming increasingly urgent for leadership.

Adding another layer: Solaris's logistics network is not performing uniformly. Its primary carrier, BlueDart, maintains a 90% OTD rate — but Delhivery, which handles over 1,200 orders, delivers on time just 52% of the time. Whether this reflects a carrier capability problem or a poor allocation of difficult orders to the wrong partner is unknown.

The Stakes

Solaris estimates that every day a late order remains undelivered costs ₹5,000 in risk-adjusted penalties (accounting for both contractual penalties and the probability of losing future contracts from affected clients). Given the concentration of late deliveries in its highest-value segment and its most strategically sensitive accounts, the true cost of inaction is far higher than the penalty number suggests.

Solaris has asked your team to go beyond the headline OTD figure, diagnose where the real vulnerabilities lie, and deliver a strategic roadmap — including a view on whether both product lines deserve equal investment going forward.

3. Data Description

Explore **Dataset**

The dataset contains 19,953 sales orders placed between November 2017 and December 2019, drawn from Solaris's ERP system. Each row represents one sales order.

Col	Field Name	Description
A	CASE_KEY	Unique identifier for each sales order.
B	DELIVERY_COMPANY	Logistics partner used to fulfil the order (BlueDart, Delhivery, FedEx, DB Schenker).
C	PRODUCT_TYPE	Product line: Titan Inverter or Eco-Battery.
D	FACTORY	Manufacturing hub that produced the order (Bengaluru, Pune, Hyderabad, Bonn, Duisburg, Wuppertal).
E	ORDERED_QUANTITY	Number of units ordered by the customer.
F	DELIVERED_QUANTITY	Number of units actually delivered. A value lower than ORDERED_QUANTITY indicates partial fulfilment.
G	MIN_ORDER_TOLERANCE	Minimum acceptable delivery quantity (lower bound of agreed tolerance window).
H	MAX_ORDER_TOLERANCE	Maximum acceptable delivery quantity (upper bound of agreed tolerance window).
I	CUST_MARKET	Customer segment: Retail, Transportation, Construction, or Other.
J	CUST_ID	Unique customer identifier.
K	CUST_NAME	Customer name (e.g., NTPC Limited, Tata Power Renewables).

L	CUST_ADDR_CODE	Customer address/region code.
M	DAYS_TO_DEL_DEADLINE	Days between PROMISED_DATE and DELIVERED_DATE. Positive = early; negative = late by that many days.
N	ORDER_TOLERANCE_MET	Binary flag (1 = Yes, 0 = No): Was the delivered quantity within the agreed tolerance window?
O	ORDER_DATE_MET	Binary flag (1 = Yes, 0 = No): Was the order delivered on or before the PROMISED_DATE? This is the primary OTD metric.
P	DELIVERED_QUANTITY_UNIT	Unit of measurement for quantity (e.g., kg).
Q	WAREHOUSE_TYPE	Type of warehouse used: Standard Hub or Smart Warehouse (automated).
R	DELIVERED_DATE	Actual date on which the order was delivered.
S	PROMISED_DATE	Date committed to the customer at the time of ordering.
T	CUST_COUNTRY	Country of the customer.
U	ORDER_VALUE	Total value of the order in INR (₹). Note: orders showing ₹0 indicate missing ERP data and can be excluded from financial calculations.
V	UNIT_PRICE	Price per unit in INR (₹).

4. Your Task

Your submission should build a consulting case that uncovers the true shape of Solaris's delivery problem, traces it to its structural roots — including a product-level dimension — and recommends a prioritised path forward. The goal is not to describe what the data shows, but to **explain what it means and what Solaris should do about it.**

Your analysis should be grounded in data-driven reasoning. Where you make causal claims or strategic recommendations, these should be supported by patterns in the data, benchmarks from analogous supply chain or logistics contexts, or well-reasoned first-principles arguments. Assertions without backing will be evaluated less favourably than conclusions clearly tied to evidence.

Act 1 — Reframe the Problem

The 87.3% headline OTD is not the right number to manage. Your first task is to disaggregate performance across factories, product lines, customer markets, and logistics partners to identify where Solaris is genuinely exposed — i.e., where OTD falls below the 80% target. For each at-risk segment, translate the gap into financial terms using the ₹5,000 daily penalty and the strategic revenue at stake. The output of this section should be a clear priority map: not all problems are equal, and your roadmap must reflect that.

Guiding question: Given that Solaris's most valuable customers are concentrated in its worst-performing segment, what is the true cost of the current performance gap?

Act 2 — Find the Drivers

With the priority areas identified, dig into the underlying causes. This section has two dimensions:

Operational diagnosis: What structural factors — factory location, warehouse type, fulfilment lead time, or logistics partner — are most correlated with delivery failure? In particular, examine whether Delhivery's OTD performance reflects an intrinsic carrier problem or whether it is being disproportionately assigned to harder-to-serve orders. **Commit to a diagnosis and defend it with data.** Where relevant, draw on established supply chain frameworks or comparable logistics case patterns to strengthen your causal reasoning.

Product lens: Titan Inverters and Eco-Battery Packs are performing very differently on OTD, fulfilment completeness, and unit economics. Analyse the two product lines across these dimensions and form a view on whether the Titan Inverter's operational problems are fixable within a reasonable investment horizon, or whether they represent a structural drag on Solaris's performance. **Your recommendation here should be commercially grounded and, where possible, informed by comparable product rationalisation decisions in the industry.**

Guiding question: Is Solaris's delivery problem a capacity issue, a carrier issue, a product issue, or some combination — and does the answer differ depending on which customer you are trying to serve?

Act 3 — Strategic Roadmap

Develop a **3-step strategic roadmap** that gives Solaris a realistic path to eliminating its sub-80% OTD segments within 12 months. Each step should be directly tied to your diagnosis in Act 2, prioritised by impact, and accompanied by a brief cost-benefit rationale.

Your roadmap must include a view on the product portfolio question: should Solaris invest equally in improving both product lines, focus its operational improvements on Eco-Battery while managing Titan Inverter volumes down, or pursue a different strategy? **This recommendation should be grounded in your Act 2 analysis and supported by a logical case for why the proposed course of action is preferable to the alternatives.**

Guiding question: If Solaris can only fix two things in the next 12 months, what are they — and why?

5. Evaluation Criteria

Your presentation will be evaluated across two dimensions:

Technical Perspective(45%)	Business Perspective(55%)
Analytical Rigour(10%) Are your findings grounded in the data, with calculations that are accurate and reproducible?	**Problem Framing(20%)** Did you correctly identify that the headline OTD number is misleading, and frame the real problem?
Design(10%) Logical sequencing, visual clarity, and cohesive presentation.	**Strategic Judgement(15%)** Does your roadmap reflect genuine prioritisation, or is it a list of everything that could be improved?
Research, Case Backing & Validation(25%) Are your strategic claims supported by data patterns, logical reasoning, or references to comparable real-world scenarios or frameworks?	**Product Recommendation(20%)** Is your view on the Titan Inverter vs Eco-Battery question commercially grounded and defensible?

6. Submission Guidelines

- Team Composition: **3–4 members per team**
- Presentation: Submit a concise deck outlining your approach and solution (**maximum 7 slides**, excluding title slide and appendix)
- Deadline: **17th April, End of Day (EOD)**
- Submission Policy: Only **one submission per team** will be accepted
- File Format: PDF file named as **<TeamName>.pdf**
- Selection Process: Shortlisted teams will be invited for **interviews** after End-Semesters
- **Content Expectation:** Slides should emphasize data-driven insights, clear problem diagnosis, and actionable strategic recommendations
- **Additional Material (Optional):** You may include supporting files such as Excel sheets, Colab notebooks, or visual assets to complement your analysis (evaluation will focus on insights, not code)

7. Resources

Business resources:

- Week 1 & 2 of Winter Consulting, C&A IITG
- Consulting Guide, C&A IITG
- Guess-it-matters - Guesstimate Guide, C&A IITG
- How to make impactful decks?

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DATASET HANDLING RESOURCES:

Steps to Follow:

1. Set Up Your Environment Use Kaggle notebooks or any other preferred environment to create graphs and analyze data.
2. Load the Dataset
   Import the dataset into your notebook for analysis
3. Explore the **Dataset**
   Get an overview of the data: check columns, data types, missing values, and summary statistics.
4. Manipulate the Data & Plot Graphs
   Clean, filter, and transform the data as needed.
   Use visualization tools to create meaningful graphs.

Draw Insights & Solve the Problem Statement Analyze the visualizations and data trends to generate insights that help address the given problem.

How to set up environment?
We recommend using Kaggle Notebooks, but you may also use Google Collab.
Here are the steps to setup your Kaggle Notebook:

- Kaggle Notebooks: A Complete Tutorial
- How to Add Kaggle Dataset to Kaggle Notebook

Python Libraries

- Basics of Python
- Python For Data Analysis
- Data Visualization with Matplotlib
- Matplotlib Kaggle Practice Notebook